
Please review and study the solutions for homework assignment 1 before proceeding with this assignment.

Homework 2 (CTL): ComS/CprE/SE 412, ComS 512

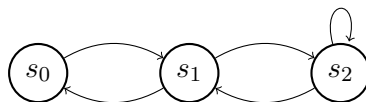
Due-date: Feb 23 at 11:59PM.

Submission format. Your submission file must be pdf format. You will use canvas-gradescope for submission (this will allow you to associate scanned pages/pdfs of your submission with the question number). Here is a short video on how to submit homework using gradescope

<https://www.youtube.com/watch?v=u-pK4GzpId0>

You are encouraged to use latex to prepare your submission but this is not a requirement.

1. Consider the following Kripke structure, with $L(s_0) = \{p\}$, $L(s_1) = \{p\}$ and $L(s_2) = \{q\}$.



Identify the set of states that satisfy each of the following and justify your answer:

- (a) $AG(p)$
- (b) $AF(p)$
- (c) $AG(AF(p))$
- (d) $AF(AG(p))$

(3+3+3+3 pts)

2. Express the following statements as CTL formula:

- (a) There exists an execution sequence such that from every configuration in the sequence, it is possible to eventually satisfy the property p .
- (b) Along all paths, whenever p is true, it is followed by a state where p is false, and whenever p is false, it is followed by a state where p is true.

(4+4 pts)

3. Prove or disprove the following:

- (a) If a state satisfies $AF(AG(p))$ then along all paths from that state the property p holds infinitely often.
- (b) If a state satisfies $AF(AG(p))$ then along all paths from that state the property $\neg p$ holds finitely many times.
- (c) If a state satisfies $EG(EF(p))$ then there exists at least one path start from that state where p holds infinitely often.
- (d) $A(p \cup AX(q))$ is equivalent to $p \wedge AX(A(p \cup q))$

- (e) *For 512. Extra credit for 412 students* The property, *for all paths of the form π , p is true in every $\pi[i]$ where i is even*, cannot be expressed in CTL. Explain in your own words why this is the case.

(3+3+3+3+4 pts)